ITL602 Web Lab – Experiment 08

Flask Currency Converter Application

1. Aim

Aim:
To build a **Python web application using Flask** and create a **simple currency converter with user input**.

2. Theory

What is Flask?

- Flask is a **Python micro web framework**.
- It helps us build **web applications using Python**.
- It is **lightweight and easy to use**.

Important Concepts

1 **Flask App**

- The main Python program that runs the web server.

2 **Route**

- URL path that defines what happens when the user visits it.
- Example: / or /convert

3 **Template**

- HTML file used to display webpages.

4 **Form**

- Used to collect input from users.

Example:

- User enters **amount**
- Selects **currency**
- Clicks **convert**

Flow of program:

User Input → Flask Server → Conversion Logic → Result

Step 1 : Create Project Folder

Open **Terminal / Command Prompt.**

Run: mkdir flask_currency_converter

This creates a folder for the project.

```
C:\Users\YASHFEEN RAAVI\Downloads>mkdir flask_currency_converter
```

Now move inside the folder:

cd flask_currency_converter

```
C:\Users\YASHFEEN RAAVI\Downloads>cd flask_currency_converter

C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter>
```

Step 2 : Create Virtual Environment

Run: python -m venv venv

```
C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter>python -m venv venv

C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter>
```

Step 3 : Activate Virtual Environment

Windows

venv\Scripts\activate

After activation you will see:

(venv) C:\Users\...

This means virtual environment is active.

```
PS C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter> venv\Scripts\activate
(venv) PS C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter>
```

Step 4 : Install Flask

Run: pip install flask

Flask will install.

You can check:

pip list

You should see **Flask** in the list.

```
(venv) PS C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter> pip install flask
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl (103 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 kB 1.5 MB/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting click>=8.1.3
  Using cached click-8.3.1-py3-none-any.whl (108 kB)
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting jinja2>=3.1.2
  Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Collecting markupsafe>=2.1.1
  Using cached markupsafe-3.0.3-cp311-cp311-win_amd64.whl (15 kB)
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.6-py3-none-any.whl (225 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 225.2/225.2 kB 3.4 MB/s eta 0:00:00
Collecting colorama
  Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: markupsafe, itsdangerous, colorama, blinker, werkzeug, jinja2, click, flask
Successfully installed blinker-1.9.0 click-8.3.1 colorama-0.4.6 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.6

[notice] A new release of pip available: 22.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter>
```

Step 5 : Create Flask Application File

Create file: app.py

Inside the project folder.

code:

```
from flask import Flask, render_template, request

app = Flask(__name__)

# Conversion rates
rates = {
   "USD_EUR": 0.85,
   "EUR_USD": 1.18,
   "USD_INR": 83,
   "INR_USD": 0.012
}

@app.route("/", methods=["GET", "POST"])
def index():
```

```
    result = None

    if request.method == "POST":
        amount = float(request.form["amount"])
        from_currency = request.form["from_currency"]
        to_currency = request.form["to_currency"]

        key = from_currency + "_" + to_currency

        if key in rates:
            result = amount * rates[key]
        else:
            result = "Conversion not available"

    return render_template("index.html", result=result)

if __name__ == "__main__":
    app.run(debug=True)
```

Step 6 : Create Templates Folder

Inside project folder create: templates

Step 7 : Create HTML Page

Inside **templates** create file:

index.html

code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Currency Converter</title>
</head>
<body>

<h2>Flask Currency Converter</h2>

<form method="POST">
```

```
Amount:
<input type="number" name="amount" required>
<br><br>

From Currency:
<select name="from_currency">
<option value="USD">USD</option>
<option value="EUR">EUR</option>
<option value="INR">INR</option>
</select>

<br><br>

To Currency:
<select name="to_currency">
<option value="USD">USD</option>
<option value="EUR">EUR</option>
<option value="INR">INR</option>
</select>

<br><br>

<input type="submit" value="Convert">

</form>

{% if result %}
<h3>Converted Amount: {{ result }}</h3>
{% endif %}

</body>
</html>
```

Step 8 : Run the Flask Application

In terminal run: python app.py

Output will show:

Running on http://127.0.0.1:5000

```
(venv) PS C:\Users\YASHFEEN RAAVI\Downloads\flask_currency_converter> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
```

Step 9 : Open in Browser

Open browser and visit: http://127.0.0.1:5000

You will see:

Flask Currency Converter

Conclusion :

The Flask Currency Converter application was successfully developed using the Flask web framework. The application accepts user input through an HTML form, processes the currency conversion using Python logic, and displays the converted result on the webpage. This experiment helped in understanding Flask routing, templates, and form handling to build a simple dynamic web application.